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Chapter 1

Introduction

1.1 Motivation

Promptable video trackers such as SAM3 [5] segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place.

1.2 Problem statement and research question

This dissertation asks whether SAM3’s zero-shot tracking accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it.

1.3 Hypotheses

H0 (null)

the label-free distances carry no predictive information about pDetA/pAssA — transfer is not distance-driven.

H1 (detection)

pDetA falls with species novelty (taxonomic + visual distance).

H2 (association)

pAssA falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 is itself an informative result — and, as Chapter 4 shows, it is what we find: neither a novelty distance nor a follow-up *intrinsic-difficulty* pivot predicts transfer beyond a trivial animal-size effect. Representational probing (does SAM3 encode taxonomy?) remains the open branch for future work.

1.4 Contributions

To our knowledge this is the first study to *predict*, before the model is run and from label-free distances, the zero-shot transfer of a promptable video tracker — and the first to split that prediction into **detection** (pDetA) and **association** (pAssA). This distinguishes it from label-free performance prediction for classifiers, which estimates a single scalar from the model’s own outputs on the target [2, 9, 15], and from the very recent static-image detector/segmenter estimators [21, 26]; and from camera-trap shift studies, which measure but do not forecast per-site reliability [3, 4, 11]. Concretely we deliver: (i) a rigorous, out-of-sample *negative* result

— under a deliberately powerful test (whole-species and whole-location hold-out with group-cluster-bootstrap significance), label-free before-running distances do *not* predict SAM 3’s transfer; (ii) the detection-versus-association decomposition, tested on two near-orthogonal splits — a constructed species hold-out (novelty) and the official location hold-out (environment); (iii) a *hardened* account of *why* it fails: a follow-up intrinsic-difficulty pivot collapses to the same animal-size confound, so the only label-free property that predicts detection is object size — the very nuisance the analysis controls for — and that positive control confirms the test detects signal when it exists; and (iv) evidence that the null is not an artefact of *one* dataset or *one* model: it holds across three datasets (SA-FARI, MammAlps, BURST) and, in a controlled model swap on BURST, across two architecturally distinct promptable trackers (GLEE and Florence-2 + SAM 2), so it is task-general rather than a quirk of SAM 3.

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.